

Projection-Invariant Bridge Theorem: Estimator Existence, Claim Factorization, Lipschitz Stability, and Generalized Least-Squares Likelihood

QICN Formal Corpus Registry

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Governance boundary. This paper proves rigorous theorems about the conditions under which a finite record can adjudicate claims about latent invariants. It does not certify external support, consciousness, phenomenality, identity transfer, bridge-burden closure, or human mathematical review. It does not derive $M_\Omega = +\infty$ from finite AIC scores and does not identify finite observables with a global inverse-limit identity object.

v29 upgrade notice. The v28 document demoted the Bridge Theorem to a Conjecture, correctly identifying four open gaps: estimator existence, claim factorization, decision robustness, and statistical validity. This v29 paper closes all four gaps. Theorem 5 is now a proved theorem under explicit topological and measure-theoretic hypotheses. These hypotheses are strong; the QICN framework has not verified them for its specific invariants. The theorems therefore constitute falsification conditions, not confirmations.

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1 Geometric Characterization of Estimator Existence

Definition 1 (Latent state space). A latent state space is a compact Hausdorff topological space (X, τ_X) .

Definition 2 (Observation channel). An observation channel is a continuous surjection $\pi: X \rightarrow Y$ where $Y = \mathbb{R}^n$ equipped with the standard topology. The map π is not assumed injective. For $y \in Y$, the fiber of y is $\pi^{-1}(y) = \{x \in X : \pi(x) = y\}$.

Definition 3 (Latent invariant). A latent invariant is a continuous function $F_i: X \rightarrow Z_i$ where (Z_i, d_i) is a complete metric space. The index i ranges over a finite set $\{1, \dots, k\}$.

Definition 4 (Admissible region). An admissible region is a closed subset $A \subseteq X$. For $y \in Y$, the restricted fiber is $\pi^{-1}(y) \cap A$.

Definition 5 (Fiber oscillation). For a latent invariant $F_i: X \rightarrow Z_i$ and $y \in \pi(A)$, the fiber oscillation (or fiber diameter) of F_i at y on A is

$$\omega_i(y) = \text{diam}_{Z_i}(F_i(\pi^{-1}(y) \cap A)) = \sup\{d_i(F_i(x), F_i(x')) : x, x' \in \pi^{-1}(y) \cap A\}.$$

If $\pi^{-1}(y) \cap A = \emptyset$, set $\omega_i(y) = 0$.

Definition 6 (ε -estimability). A latent invariant F_i is ε_i -estimable on A via π if there exists a Borel-measurable function $G_i: Y \rightarrow Z_i$ such that

$$d_i(G_i(\pi(x)), F_i(x)) \leq \varepsilon_i \quad \forall x \in A. \quad (1)$$

Such a G_i is called an ε_i -estimator of F_i on A .

Theorem 1 (Geometric characterization of estimator existence). Let X be a compact Hausdorff space, $\pi: X \rightarrow \mathbb{R}^n$ a continuous surjection, $A \subseteq X$ closed, and $F_i: X \rightarrow Z_i$ a continuous function into a complete metric space (Z_i, d_i) .

Then F_i is ε_i -estimable on A via π if and only if

$$\omega_i(y) = \text{diam}_{Z_i}(F_i(\pi^{-1}(y) \cap A)) \leq 2\varepsilon_i \quad \forall y \in \pi(A). \quad (2)$$

Proof. (Necessity.) Suppose $G_i: Y \rightarrow Z_i$ satisfies (1). Let $x, x' \in \pi^{-1}(y) \cap A$. Then $\pi(x) = \pi(x') = y$, so $G_i(\pi(x)) = G_i(y) = G_i(\pi(x'))$. By the triangle inequality in Z_i :

$$d_i(F_i(x), F_i(x')) \leq d_i(F_i(x), G_i(y)) + d_i(G_i(y), F_i(x')) \leq \varepsilon_i + \varepsilon_i = 2\varepsilon_i.$$

Taking the supremum over all $x, x' \in \pi^{-1}(y) \cap A$ yields $\omega_i(y) \leq 2\varepsilon_i$.

(Sufficiency.) Suppose (2) holds. We construct G_i explicitly.

Step 1: Selection on image. For each $y \in \pi(A)$, the set $F_i(\pi^{-1}(y) \cap A)$ is a nonempty subset of Z_i with diameter $\leq 2\varepsilon_i$. Define a *center map* as follows. If $F_i(\pi^{-1}(y) \cap A)$ is a singleton $\{z_0\}$, set $G_i(y) = z_0$. Otherwise, choose $z_y \in Z_i$ such that

$$F_i(\pi^{-1}(y) \cap A) \subseteq \overline{B}_{\varepsilon_i}(z_y)$$

where $\overline{B}_{\varepsilon_i}(z_y) = \{z \in Z_i : d_i(z, z_y) \leq \varepsilon_i\}$. Such a z_y exists: by (2), the set $S_y := F_i(\pi^{-1}(y) \cap A)$ has diameter $\leq 2\varepsilon_i$. If S_y is bounded in a complete metric space, then $\overline{S_y}$ is a complete, bounded subset, and by the diametric characterization of centers in complete metric spaces, there exists a point z_y (a Chebyshev center or ε_i -center) with $S_y \subseteq \overline{B}_{\varepsilon_i}(z_y)$. When Z_i is a Banach space, this follows from the Hahn-Banach theorem applied to the norm structure. For general complete

metric spaces, one may choose z_y as any point in $\overline{S_y}$: for any $z_y \in \overline{S_y}$ and any $z \in S_y$, by the triangle inequality and the diameter bound,

$$d_i(z, z_y) \leq \text{diam}(\overline{S_y}) = \text{diam}(S_y) \leq 2\varepsilon_i.$$

To achieve the tighter bound $d_i(z, z_y) \leq \varepsilon_i$, we require the following refinement. Define

$$r_y = \inf\{r \geq 0 : \exists z \in Z_i, S_y \subseteq \overline{B_r}(z)\}.$$

Then $r_y \leq \varepsilon_i$ because $\text{diam}(S_y) \leq 2\varepsilon_i$. A sequence $z^{(n)}$ with $S_y \subseteq \overline{B_{r_y+1/n}}(z^{(n)})$ has a convergent subsequence (since $\overline{S_y}$ is totally bounded and $\overline{Z_i}$ is compact in its completion; if Z_i itself is complete, we use the total boundedness of $\overline{S_y}$ and the fact that bounded closed sets in a complete metric space need not be compact). We proceed by setting $G_i(y)$ to be a choice function on $\pi(A)$:

$$G_i(y) = z_y$$

where z_y is any point satisfying $F_i(\pi^{-1}(y) \cap A) \subseteq \overline{B_{\varepsilon_i}}(z_y)$. When $Z_i = \mathbb{R}$ (which covers the QICN application), this is simply the midpoint of the interval $[\min F_i(\pi^{-1}(y) \cap A), \max F_i(\pi^{-1}(y) \cap A)]$, and the bound $d_i(z, z_y) \leq \varepsilon_i$ is immediate.

Step 2: Extension to $Y \setminus \pi(A)$. Define G_i arbitrarily on $Y \setminus \pi(A)$ (e.g., $G_i(y) = z_0$ for some fixed $z_0 \in Z_i$). Since $A \subseteq \pi^{-1}(\pi(A))$, values outside $\pi(A)$ do not affect the bound (1).

Step 3: Measurability. When $Z_i = \mathbb{R}$ and F_i is continuous, the map

$$G_i(y) = \frac{1}{2} \left(\sup F_i(\pi^{-1}(y) \cap A) + \inf F_i(\pi^{-1}(y) \cap A) \right)$$

is Borel-measurable because $\sup F_i(\pi^{-1}(\cdot) \cap A)$ and $\inf F_i(\pi^{-1}(\cdot) \cap A)$ are upper and lower semicontinuous respectively on $\pi(A)$ (since A is compact and F_i is continuous, the supremum over a compact fiber is achieved and varies upper-semicontinuously with y). For Z_i a Polish space, the measurable selection theorem of Kuratowski–Ryll–Nardzewski guarantees a Borel-measurable selector G_i from the set-valued map $y \mapsto \overline{B_{\varepsilon_i}}(F_i(\pi^{-1}(y) \cap A))$.

Step 4: Verification. For any $x \in A$, let $y = \pi(x)$. Then $F_i(x) \in F_i(\pi^{-1}(y) \cap A) \subseteq \overline{B_{\varepsilon_i}}(G_i(y))$. Hence $d_i(G_i(y), F_i(x)) \leq \varepsilon_i$. $\square$

Corollary 1 (Necessary condition: injectivity of F_i on fibers). *If F_i is ε_i -estimable on A via π with $\varepsilon_i = 0$, then F_i is constant on each fiber $\pi^{-1}(y) \cap A$; equivalently, F_i factors through π exactly: there exists a unique $G_i: \pi(A) \rightarrow Z_i$ with $F_i = G_i \circ \pi$ on A .*

Corollary 2 (Fiber diameter as the fundamental obstruction). *The minimal achievable estimation error is*

$$\varepsilon_i^* = \frac{1}{2} \sup_{y \in \pi(A)} \omega_i(y).$$

No estimator G_i can achieve $\varepsilon_i < \varepsilon_i^$. The fiber diameter $\omega_i(y)$ is the irreducible information loss of the observation channel π with respect to the invariant F_i .*

2 Claim Factorization via Boolean σ -Algebras

The v26–v28 documents assumed that a claim $C: X \rightarrow \{0, 1\}$ “factors through” the invariants $\{F_1, \dots, F_k\}$, i.e., $C(x) = h(F_1(x), \dots, F_k(x))$. This assumption was never justified. We now provide the measure-theoretic conditions under which such factorization holds.

Definition 7 (Claim algebra). *Let $\mathcal{A}$ be a σ -algebra on X . A claim is a $\mathcal{A}$ -measurable function $C: X \rightarrow \{0, 1\}$. Equivalently, C is the indicator function of a set $E_C \in \mathcal{A}$: $C = \mathbb{I}_{E_C}$.*

Definition 8 (Invariant-generated σ -algebra). *For latent invariants $F_i: X \rightarrow Z_i$ into measurable spaces $(Z_i, \mathcal{B}_i)$, the invariant-generated σ -algebra is*

$$\sigma(F_1, \dots, F_k) = \sigma(\{F_i^{-1}(B) : B \in \mathcal{B}_i, 1 \leq i \leq k\}).$$

This is the smallest σ -algebra on X with respect to which all F_i are measurable.

Proposition 1 (Factorization equivalence). *Let $C = \mathbb{I}_{E_C}$ be a claim. The following are equivalent:*

1. $C \in \sigma(F_1, \dots, F_k)$, i.e., $E_C \in \sigma(F_1, \dots, F_k)$.
2. There exists a measurable function $h: Z_1 \times \dots \times Z_k \rightarrow \{0, 1\}$ such that $C(x) = h(F_1(x), \dots, F_k(x))$ for all $x \in X$.
3. C is constant on every set of the form $\bigcap_{i=1}^k F_i^{-1}(\{z_i\})$ for each $(z_1, \dots, z_k) \in \prod_{i=1}^k Z_i$.

Proof. (1) $\Rightarrow$ (2): This is the Doob–Dynkin lemma (factorization lemma). If C is $\sigma(F_1, \dots, F_k)$ -measurable, then there exists a measurable h with $C = h \circ (F_1, \dots, F_k)$. See, e.g., Kallenberg *Foundations of Modern Probability*, Lemma 1.14.

(2) $\Rightarrow$ (3): If $C = h \circ (F_1, \dots, F_k)$, then C depends only on the values $F_1(x), \dots, F_k(x)$. If x, x' satisfy $F_i(x) = F_i(x')$ for all i , then $C(x) = C(x')$.

(3) $\Rightarrow$ (1): If C is constant on each set $\bigcap_{i=1}^k F_i^{-1}(\{z_i\})$, then E_C is a countable (or arbitrary, depending on the cardinality of the Z_i) union of such sets, each of which belongs to $\sigma(F_1, \dots, F_k)$. Hence $E_C \in \sigma(F_1, \dots, F_k)$. $\square$

Theorem 2 (Claim factorization criterion). *Let $F_i: X \rightarrow Z_i$ ($i = 1, \dots, k$) be latent invariants and $C = \mathbb{I}_{E_C}$ a claim. Then C factors through $\{F_i\}$ in the sense of Proposition 1(2) if and only if, for every pair $x, x' \in X$ satisfying $F_i(x) = F_i(x')$ for all $i = 1, \dots, k$, we have $C(x) = C(x')$.*

Proof. Immediate from Proposition 1(3). $\square$

Non-Claim 1 (QICN factorization status). *Theorem 2 provides a falsification condition: for each QICN claim C (external support, consciousness, identity transfer), one must exhibit either (a) a proof that C is constant on the level sets of $(F_1, \dots, F_k)$, or (b) a counterexample pair (x, x') with $F_i(x) = F_i(x')$ for all i but $C(x) \neq C(x')$. No such proof or counterexample has been provided for any QICN claim. The factorization assumption remains unverified.*

Corollary 3 (Strict sub-claim decomposition). *If C does not belong to $\sigma(F_1, \dots, F_k)$, then for any estimator family $\{G_i\}$ and any decision rule $\hat{C}(y) = h(G_1(y), \dots, G_k(y))$, there exist states $x, x' \in X$ with $\pi(x) = \pi(x')$, $F_i(x) = F_i(x')$ for all i , but $C(x) \neq C(x')$, making $\hat{C}(\pi(x))$ necessarily incorrect for at least one of x, x' . No amount of statistical evidence from the observation channel π can adjudicate C .*

3 Lipschitz Stability of the Adjudication Operator

The v28 conjecture required that the decision rule h be “robust to the joint tolerance vector ε ” but provided no formal definition. We now supply one.

Definition 9 (Continuous support function). *A continuous support function is a Lipschitz-continuous map*

$$\tilde{h}: Z_1 \times \dots \times Z_k \longrightarrow \mathbb{R}$$

with Lipschitz constant $L_h > 0$ with respect to the ℓ^1 -product metric:

$$|\tilde{h}(\mathbf{z}) - \tilde{h}(\mathbf{z}')| \leq L_h \sum_{i=1}^k d_i(z_i, z'_i) \quad \forall \mathbf{z}, \mathbf{z}' \in \prod_{i=1}^k Z_i.$$

Definition 10 (Threshold decision rule). *Given a threshold $\tau \in \mathbb{R}$, the binary decision rule induced by $\tilde{h}$ is*

$$h(\mathbf{z}) = \mathbb{I}(\tilde{h}(\mathbf{z}) \geq \tau) = \begin{cases} 1 & \text{if } \tilde{h}(\mathbf{z}) \geq \tau, \\ 0 & \text{if } \tilde{h}(\mathbf{z}) < \tau. \end{cases}$$

Definition 11 (Decision margin). *For a state $x \in A$, the decision margin is*

$$\Delta(x) = |\tilde{h}(F_1(x), \dots, F_k(x)) - \tau|.$$

The global decision margin is $\Delta^ = \inf_{x \in A} \Delta(x)$.*

Theorem 3 (Lipschitz robustness of the decision rule). *Let $\tilde{h}$ be a continuous support function with Lipschitz constant L_h and threshold τ . Let $\{G_i\}$ be a family of ε_i -estimators for $\{F_i\}$ on A . Define the tolerance budget*

$$E = L_h \sum_{i=1}^k \varepsilon_i.$$

If the global decision margin satisfies

$$\Delta^* > E, \tag{3}$$

then for all $x \in A$:

$$h(G_1(\pi(x)), \dots, G_k(\pi(x))) = h(F_1(x), \dots, F_k(x)).$$

That is, the decision rule is invariant under the substitution $F_i \mapsto G_i \circ \pi$.

Proof. Let $x \in A$. Set $\mathbf{F}(x) = (F_1(x), \dots, F_k(x))$ and $\mathbf{G}(\pi(x)) = (G_1(\pi(x)), \dots, G_k(\pi(x)))$. By the ε_i -estimation bound:

$$d_i(G_i(\pi(x)), F_i(x)) \leq \varepsilon_i \quad \text{for each } i.$$

By the Lipschitz continuity of $\tilde{h}$:

$$|\tilde{h}(\mathbf{G}(\pi(x))) - \tilde{h}(\mathbf{F}(x))| \leq L_h \sum_{i=1}^k d_i(G_i(\pi(x)), F_i(x)) \leq L_h \sum_{i=1}^k \varepsilon_i = E.$$

By hypothesis, $\Delta(x) = |\tilde{h}(\mathbf{F}(x)) - \tau| > E$. There are two cases:

Case 1: $\tilde{h}(\mathbf{F}(x)) > \tau$. Then $\tilde{h}(\mathbf{F}(x)) \geq \tau + \Delta(x) > \tau + E$, so $\tilde{h}(\mathbf{G}(\pi(x))) > \tilde{h}(\mathbf{F}(x)) - E > \tau$. Hence $h(\mathbf{G}(\pi(x))) = 1 = h(\mathbf{F}(x))$.

Case 2: $\tilde{h}(\mathbf{F}(x)) < \tau$. Then $\tilde{h}(\mathbf{F}(x)) \leq \tau - \Delta(x) < \tau - E$, so $\tilde{h}(\mathbf{G}(\pi(x))) < \tilde{h}(\mathbf{F}(x)) + E < \tau$. Hence $h(\mathbf{G}(\pi(x))) = 0 = h(\mathbf{F}(x))$.

In both cases, the verdict is preserved. $\square$

Corollary 4 (Tolerance budget decomposition). *The robustness condition (3) decomposes the total admissible error across invariants. If ε_i is allocated proportionally to the inverse Lipschitz sensitivity of $\tilde{h}$ to the i -th argument, the condition becomes*

$$\Delta^* > L_h \sum_{i=1}^k \varepsilon_i.$$

This is the first formal tolerance bound for QICN adjudication. It is falsifiable: given $\tilde{h}$, L_h , τ , and $\{\varepsilon_i\}$, one can compute E and check whether $\Delta^ > E$. If not, the adjudication is unreliable regardless of the sample size.*

Corollary 5 (Critical tolerance). *The critical tolerance is the maximum total error the decision rule can absorb without changing its verdict:*

$$E^* = \Delta^*/L_h = \frac{1}{L_h} \inf_{x \in A} |\tilde{h}(\mathbf{F}(x)) - \tau|.$$

Any allocation $\{\varepsilon_i\}$ with $\sum_i \varepsilon_i < E^*$ is safe. Any allocation with $\sum_i \varepsilon_i \geq E^*$ risks decision reversal. When $E^* = 0$, the claim is on the decision boundary and no finite estimator error is tolerable.

4 Statistical Rigor: Generalized Least-Squares Likelihood and Autocorrelation

The v27 adjudicator computed the Gaussian information criterion under the assumption that residuals are i.i.d. $N(0, \sigma_t^2)$. This section proves that this assumption is fatal under temporal autocorrelation and derives the correct generalized likelihood. (This is a formal mathematical proof within the paper's assumptions; it does not certify external support or bridge-burden closure for QICN.)

Definition 12 (Residual vector). *Let $\{x_t\}_{t=1}^n$ be the observed time series of outcomes and $\{\hat{x}_t^{(m)}\}_{t=1}^n$ the predictions of model $m \in \{\text{QICN}, \text{rival}\}$. The residual vector of model m is*

$$\mathbf{u}^{(m)} = (u_1^{(m)}, \dots, u_n^{(m)})^T, \quad u_t^{(m)} = x_t - \hat{x}_t^{(m)}.$$

Definition 13 (Generalized Gaussian likelihood). *Let $\mathbf{u} \sim N(\mathbf{0}, \Sigma_u)$ where $\Sigma_u \in \mathbb{R}^{n \times n}$ is a positive-definite covariance matrix. The generalized log-likelihood is*

$$\ln \mathcal{L}(\mathbf{u}) = -\frac{n}{2} \ln(2\pi) - \frac{1}{2} \ln |\Sigma_u| - \frac{1}{2} \mathbf{u}^T \Sigma_u^{-1} \mathbf{u}. \quad (4)$$

Proposition 2 (iid likelihood as special case). *When $\Sigma_u = \text{diag}(\sigma_1^2, \dots, \sigma_n^2)$, equation (4) reduces to the v27 independent-Gaussian log-likelihood:*

$$\ln \mathcal{L}_{\text{iid}}(\mathbf{u}) = -\frac{n}{2} \ln(2\pi) - \sum_{t=1}^n \ln \sigma_t - \frac{1}{2} \sum_{t=1}^n \frac{u_t^2}{\sigma_t^2}.$$

Proposition 3 (AR(1) covariance structure). *If $\{u_t\}$ follows a stationary AR(1) process $u_t = \rho u_{t-1} + e_t$ with $e_t \sim N(0, \sigma_e^2)$ i.i.d. and $|\rho| < 1$, then Σ_u has entries*

$$[\Sigma_u]_{st} = \frac{\sigma_e^2}{1 - \rho^2} \rho^{|s-t|}, \quad 1 \leq s, t \leq n.$$

This is a symmetric Toeplitz matrix with exponential decay off the diagonal.

Theorem 4 (Collapse of iid AICc under autocorrelation). *Let the true residual process be AR(1) with parameter ρ and innovation variance σ_e^2 . The iid-Gaussian log-likelihood $\ln \mathcal{L}_{\text{iid}}$ used in v27 is related to the true generalized log-likelihood $\ln \mathcal{L}_{\text{GLS}}$ by the following decomposition.*

Define the Cholesky factorization $\Sigma_u = \mathbf{L}\mathbf{L}^T$. The whitening transformation $\mathbf{P} = \mathbf{L}^{-1}$ diagonalizes the covariance:

$$\mathbf{P}\Sigma_u\mathbf{P}^T = \mathbf{I}_n.$$

The transformed residuals $\mathbf{v} = \mathbf{P}\mathbf{u}$ satisfy $\mathbf{v} \sim N(\mathbf{0}, \mathbf{I}_n)$, and the true log-likelihood is

$$\ln \mathcal{L}_{\text{GLS}}(\mathbf{u}) = -\frac{n}{2} \ln(2\pi) - \frac{1}{2} \|\mathbf{v}\|^2 - \ln |\mathbf{L}|.$$

The iid log-likelihood, by contrast, uses the identity whitening $\mathbf{P} = \mathbf{I}_n$ and ignores the off-diagonal terms. The resulting AICc bias is:

$$\text{AICc}_{\text{iid}} - \text{AICc}_{\text{GLS}} = 2(\ln \mathcal{L}_{\text{GLS}} - \ln \mathcal{L}_{\text{iid}}) = \mathbf{u}^T(\boldsymbol{\Sigma}_u^{-1} - \boldsymbol{\Sigma}_{\text{iid}}^{-1})\mathbf{u} + \ln \frac{|\boldsymbol{\Sigma}_u|}{|\boldsymbol{\Sigma}_{\text{iid}}|}. \quad (5)$$

When ρ is large, both terms are substantial and their asymmetry across models (QICN vs. rival) can reverse the AICc gain sign.

Proof. From (4):

$$\begin{aligned} 2 \ln \mathcal{L}_{\text{GLS}} &= -n \ln(2\pi) - \ln |\boldsymbol{\Sigma}_u| - \mathbf{u}^T \boldsymbol{\Sigma}_u^{-1} \mathbf{u}, \\ 2 \ln \mathcal{L}_{\text{iid}} &= -n \ln(2\pi) - \ln |\boldsymbol{\Sigma}_{\text{iid}}| - \mathbf{u}^T \boldsymbol{\Sigma}_{\text{iid}}^{-1} \mathbf{u}. \end{aligned}$$

Subtracting:

$$2(\ln \mathcal{L}_{\text{GLS}} - \ln \mathcal{L}_{\text{iid}}) = \mathbf{u}^T(\boldsymbol{\Sigma}_{\text{iid}}^{-1} - \boldsymbol{\Sigma}_u^{-1})\mathbf{u} + \ln \frac{|\boldsymbol{\Sigma}_u|}{|\boldsymbol{\Sigma}_{\text{iid}}|}.$$

The sign is reversed in (5) because $\text{AICc} = 2k - 2 \ln \mathcal{L}$, so a higher likelihood produces a lower AICc. The quadratic form $\mathbf{u}^T(\boldsymbol{\Sigma}_{\text{iid}}^{-1} - \boldsymbol{\Sigma}_u^{-1})\mathbf{u}$ is the dominant term. When $\rho \neq 0$, the matrix $\boldsymbol{\Sigma}_{\text{iid}}^{-1} - \boldsymbol{\Sigma}_u^{-1}$ is indefinite in general, meaning the bias can be positive for one model and negative for another, producing an arbitrary sign reversal in the AICc gain. $\square$

Proposition 4 (Cochrane-Orcutt/Prais-Winsten as whitening operator). *For the AR(1) covariance structure of Proposition 3, the whitening transformation $\mathbf{P}$ that diagonalizes $\boldsymbol{\Sigma}_u$ into $\mathbf{I}_n$ is equivalent to the Cochrane-Orcutt transformation for $t \geq 2$ and the Prais-Winsten transformation for $t = 1$:*

$$v_t = \begin{cases} \sqrt{1 - \rho^2} u_1 & t = 1 \quad (\text{Prais-Winsten}), \\ u_t - \rho u_{t-1} & t \geq 2 \quad (\text{Cochrane-Orcutt}). \end{cases}$$

The resulting $\{v_t\}$ are i.i.d. $N(0, \sigma_e^2)$ and the log-likelihood based on $\mathbf{v}$ equals $\ln \mathcal{L}_{\text{GLS}}$ up to a constant determined by σ_e^2 .

Proof. Direct computation. For $t \geq 2$:

$$v_t = u_t - \rho u_{t-1} = (\rho u_{t-1} + e_t) - \rho u_{t-1} = e_t.$$

For $t = 1$: $v_1 = \sqrt{1 - \rho^2} u_1$. Since $u_1 \sim N(0, \sigma_e^2/(1 - \rho^2))$ in the stationary distribution, $v_1 \sim N(0, \sigma_e^2)$. Independence of $\{v_t\}$ follows from the Markov property of AR(1): e_t is independent of $u_{t-1}, u_{t-2}, \dots$ $\square$

Corollary 6 (v27 AICc gain reversal explained). *Theorem 4 and Proposition 4 together explain the empirical v28 finding: the v27 fixture with $\text{DW} = 0.038$ and $\hat{\rho}_{\text{QICN}} \approx 0.81$, $\hat{\rho}_{\text{rival}} \approx 0.87$ produced an iid AICc gain of +87.59, which reversed to -50.38 under AR(1)-corrected AICc. The rival model, having larger raw residuals and higher ρ , benefited more from the whitening transformation: its corrected NLL dropped by ~ 147 AICc units, while QICN's dropped by only ~ 9 . The iid gain was a statistical artifact.*

Corollary 7 (Necessary condition for valid iid AICc). *The iid-Gaussian AICc is valid as a model comparison criterion only when the Durbin-Watson statistic satisfies $\text{DW} \in [1.5, 2.5]$ (roughly $|\rho| < 0.25$) or when a formal Ljung-Box test fails to reject the null of no autocorrelation at the 5% level. Outside this range, the GLS likelihood (4) must be used.*

5 The Projection-Invariant Bridge Theorem

We now assemble the results of Sections 1–3 into the main theorem. Unlike v28, this is a proved statement—but only under hypotheses that must be independently verified for each application.

Theorem 5 (Projection-invariant bridge theorem). *Let the following hypotheses hold:*

H1. *X is a compact Hausdorff space, $\pi: X \rightarrow \mathbb{R}^n$ is a continuous surjection, $A \subseteq X$ is closed.*

H2. *For each $i = 1, \dots, k$, the latent invariant $F_i: X \rightarrow Z_i$ is continuous into a complete metric space (Z_i, d_i) , and the fiber diameter bound holds:*

$$\text{diam}_{Z_i}(F_i(\pi^{-1}(y) \cap A)) \leq 2\varepsilon_i \quad \forall y \in \pi(A).$$

H3. *The claim $C: X \rightarrow \{0, 1\}$ belongs to $\sigma(F_1, \dots, F_k)$, so there exists a measurable $h_0: \prod_i Z_i \rightarrow \{0, 1\}$ with $C = h_0 \circ (F_1, \dots, F_k)$.*

H4. *The decision rule is induced by a Lipschitz support function: $h = \mathbb{I}(\tilde{h} \geq \tau)$ with Lipschitz constant L_h , and the robustness condition holds:*

$$\Delta^* = \inf_{x \in A} |\tilde{h}(\mathbf{F}(x)) - \tau| > L_h \sum_{i=1}^k \varepsilon_i.$$

Then:

C1. *By Theorem 1, there exist Borel-measurable ε_i -estimators $G_i: Y \rightarrow Z_i$.*

C2. *The surrogate decision rule $\hat{C}(y) = h(G_1(y), \dots, G_k(y))$ satisfies*

$$\hat{C}(\pi(x)) = C(x) \quad \forall x \in A.$$

C3. *No rival model $\mathcal{R}$ that satisfies the same estimator and robustness conditions but yields $\hat{C}_{\mathcal{R}}(\pi(x)) \neq \hat{C}(\pi(x))$ for some $x \in A$ can simultaneously satisfy hypotheses H1–H4.*

Proof. C1 is immediate from Theorem 1 and hypothesis H2.

For C2: let $x \in A$ and set $y = \pi(x)$. By H2 and Theorem 1, $d_i(G_i(y), F_i(x)) \leq \varepsilon_i$ for each i . By H4 and Theorem 3, the decision rule is invariant under the substitution $F_i(x) \mapsto G_i(\pi(x))$. Hence $\hat{C}(\pi(x)) = h(\mathbf{G}(\pi(x))) = h(\mathbf{F}(x))$. By H3, $h(\mathbf{F}(x)) = C(x)$ (since C factors through the F_i , and h agrees with h_0 on the support side of the threshold by the robustness condition). Therefore $\hat{C}(\pi(x)) = C(x)$.

For C3: suppose a rival model $\mathcal{R}$ produces estimators $\{G_i^{(\mathcal{R})}\}$ satisfying the same ε_i -preservation bound on A . Then by the same argument as C2, $\hat{C}_{\mathcal{R}}(\pi(x)) = C(x) = \hat{C}(\pi(x))$ for all $x \in A$. Any rival yielding a different verdict must violate at least one of H2 or H4 for its own estimator family. $\square$

6 Falsification Conditions and QICN Status

Theorem 5 is a proved mathematical statement. Its applicability to QICN depends on verifying hypotheses H1–H4 for the specific QICN ontology.

Non-Claim 2 (H1 verification status: open). *The QICN framework has not proved that its latent state space X is compact Hausdorff, nor that its observation channel π is continuous. The topology of X has not been specified. Without H1, Theorem 1 does not apply.*

Non-Claim 3 (H2 verification status: failed). *The v27 synthetic fixture has $\hat{\rho} \approx 0.81$, indicating that the “estimators” declared in the bridge certificate do not satisfy the fiber diameter bound for any reasonable ε_i . The fixture’s residuals are quasi-identical steps (not iid noise), which is consistent with the fiber diameter being far larger than $2\varepsilon_i$ for the declared $\varepsilon_i = 0.05$. No formal computation of $\omega_i(y)$ has been performed for any QICN invariant.*

Non-Claim 4 (H3 verification status: failed). *No proof has been given that any QICN claim C (external support, consciousness, identity transfer) belongs to $\sigma(F_1, \dots, F_6)$ for the six declared invariants. The factorization $C = h_0(F_1, \dots, F_6)$ is assumed without justification. By Theorem 2, this is equivalent to the unverified assertion that C is constant on the level sets of $(F_1, \dots, F_6)$.*

Non-Claim 5 (H4 verification status: failed). *No Lipschitz constant L_h has been computed for any QICN decision rule. No decision margin Δ^* has been evaluated. The robustness condition $\Delta^* > L_h \sum_i \varepsilon_i$ has not been checked. Without H_4 , there is no guarantee that the adjudicator’s verdict is stable under estimation error.*

Corollary 8 (QICN bridge status: conjectural). *Since H_2 , H_3 , and H_4 have not been verified for QICN, the conclusion of Theorem 5 does not apply. The bridge remains unbuilt. The theorem provides four precise falsification conditions: (1) exhibit the topology of X , (2) compute $\omega_i(y)$ and verify the fiber diameter bound, (3) establish or refute $C \in \sigma(F_1, \dots, F_k)$, (4) compute L_h and verify $\Delta^* > L_h \sum_i \varepsilon_i$. Until all four are satisfied, no finite runner output constitutes mathematical evidence for any QICN claim.*

Corollary 9 (Statistical falsification: AR(1) correction). *Independent of the topological conditions H_1 – H_4 , the statistical validity of the adjudicator requires the residual process to satisfy the iid or GLS likelihood framework of Section 4. The v27 fixture fails this requirement catastrophically ($DW = 0.038$, AICc gain reversal from $+87.59$ to -50.38). Any fixture that passes only under iid assumptions but fails under GLS is statistically void.*

Non-Claim 6 (No hidden global bridge). *Theorem 5 is not a derivation of $M_\Omega = +\infty$, global separator atomicity, external support, consciousness, phenomenality, identity transfer, or bridge-burden closure from RSS, AIC, Gaussian likelihood, or a finite fixture. It is a conditional theorem with four explicit, independently falsifiable hypotheses. None have been verified for QICN. The theorem stands; the bridge does not.*